

Total No. of Questions : 8]

SEAT No. :

P646

[Total No. of Pages : 2

[5869]-273

S.E. (Computer/Artificial Intelligence & Data Science)

OBJECT ORIENTED PROGRAMMING

(2019 Pattern) (Semester - III) (Theory) (210243)

Time : 2.½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Draw neat and clean diagram.
- 3) Assume suitable data, if Necessary.

- Q1) a)** Differentiate between compile time polymorphism and runtime Polymorphism. along with examples. **[5]**
- b) Explain the use of mutual and explicit keywords in C++ language along with example. **[6]**
- c) Write a program to demonstrate for overloading decrement[--] Operator in C++. (Assume a class number with data member as int x) **[6]**

OR

- Q2) a)** Explain the concept of method/function overriding along with examples. What is operator overloading in polymorphism? Write the program to overload-operator for subtracting two complex numbers which are object of below complex class by defining operator function in below. **[5]**
- b) Class.
- Class complex{
- Private:int real, imag; }; **[6]**
- c) What is virtual function and its use? How we can access virtual functions through the use of pointer to the base class. **[6]**

- Q3) a)** What are cin and cout? Explain iostream. **[4]**
- b) Write a program that returns the size in bytes of a program entered on the command line. **[6]**
- c) What is a stream? Write a program to illustrate the stream errors. **[8]**

OR

P.T.O.

- Q4)** a) What is a file mode? Describe the various file mode options available.[4]
b) Write a program to create files using constructor function. [6]
c) Write a program using C++ file input and output class with open(), get(), close () for opening, reading from and writing to a file. [8]

- Q5)** a) What is need of exception handling? Explain types of exception. [4]
b) What is template? Write a program to handle addition of two numbers.[6]
c) List and explain four common examples of exceptions. [7]

OR

- Q6)** a) What is difference between class template and function template in C++? [4]
b) Write a function template for finding the minimum value contained in an array [6]
c) What are user-defined exceptions? Explain with suitable example? [7]

- Q7)** a) List and explain different types of iterators in STL. [4]
b) What is Stack? How is it implemented in STL. [6]
c) Use minimum 8 functions of Deque STL. Write a program to explain the same. [8]

OR

- Q8)** a) Elaborate Advantages and Disadvantages of LIST Sequential Container.[4]
b) What is meant by Associative container? State and explain the types.[6]
c) Use minimum 8 functions of vector STL. Write a program to explain the same. [8]

